



Development and validation of ChatGPT literacy scale

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Abstract

ChatGPT revolutionized the landscape of information transitions and creativity. The number of ChatGPT users is increasing at the fastest pace in consumer applications history, which calls for a reliable ChatGPT literacy scale. To that end, first, we conducted a literature review to gain a conceptual understanding of the ChatGPT literacy construct. Next, we conducted a focus group consisting of five experts to brainstorm potential items and the following five factors of ChatGPT literacy: technical proficiency, critical evaluation, communication proficiency, creative application, and ethical competence. The Delphi method with 10 experts was used to validate these items. After collecting expert feedback and evaluation data, we calculated content validity ratios to analyze the items' content validity. Another focus group interview was conducted with 10 university students as a pilot test. Finally, 822 college students were recruited to evaluate the construct validity of ChatGPT literacy through exploratory and confirmatory factor analysis, resulting in 25 final items. This novel approach is expected to contribute to building a theoretical basis for ChatGPT and large language model studies, providing practical implications for large language model developers, helping to develop related regulations and policies, and creating a healthy digital culture for the use of ChatGPT.

Keywords ChatGPT · Digital literacy · AI literacy · Scale development · Delphi · Validity

ChatGPT, an artificial intelligence (AI)-based language model, is revolutionizing the landscape of information transition and creativity (Gordijn & Have, 2023). It has seen the fastest growth in the history of consumer applications. It took two months for ChatGPT to reach 100 million users, the same number Instagram had in 2.5 years (Clay, 2023; Homolak, 2023; Reed, 2023). The continuous evolution of ChatGPT and its ability to converse with users almost authentically have prompted both excitement and concern in various fields, such as medicine, healthcare, and education (Anderson et al., 2023; Cascella et al., 2023; Hachman, 2022). Experts have pointed out possible problems such as

misinformation, information ownership, and information bias, and developers, governments, and organizations are actively trying to solve these problems (Chan, 2022; Rosenblatt, 2023; Weinstein, 2023). However, we argue that technical development and regulations alone cannot solve the problems and questions surrounding ChatGPT; a user-level digital culture change is required to make ChatGPT a revolution rather than a disaster (Chan, 2022; Floridi & Chiriatti, 2020).

To create a healthy digital culture surrounding the use of ChatGPT, it is necessary to measure the ChatGPT literacy of individual users accurately. Previous studies have highlighted the importance of digital literacy (e.g., new media, AI, and computer literacy) as a basic life skill and an essential communication and education tool (Bawden, 2008; Eshet, 2012; Koc & Barut, 2016; Ng et al., 2021; Sarkar, 2012; Wang et al., 2022). Digital literacy should be considered not only a necessary tool but also a key virtue for citizens of this user-centered digital era, encompassing basic knowledge and technical skills in certain technologies as well as the critical evaluation and ethical competence surrounding the use of technologies (Eshet, 2012; Gilster, 1997; Floridi & Chiriatti, 2020). This will become even more true

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as technological development furthers democratization and a coactive digital environment (Gilster, 1997; Raschke, 2003).

Based on the discussion, we argue that the development of an accurate ChatGPT literacy scale is necessary considering the rapid growth of users and its social impact. Therefore, this study attempts to answer the following question: What factors and items constitute a valid and reliable ChatGPT literacy scale? To this end, this study first conducted a literature review to extract key factors and potential items. A focus group interview with experts and the Delphi method were employed for content validation. A pilot study was conducted on face validity, and an online survey was conducted to evaluate the construct, convergent, and discriminant validity of the ChatGPT literacy scale through exploratory and confirmatory factor analyses.

By answering the question (i.e., What factors and items constitute a valid and reliable ChatGPT literacy scale?), we expect this study to provide a basis for future studies exploring the role of ChatGPT literacy skills in education, business, creative work, etc. Moreover, the ChatGPT literacy scale can be utilized by educational institutions to examine students' ability to use ChatGPT properly and ethically as well as the relationship between the ability and various educational outcomes such as achievements and student satisfaction. We also believe that the scale can be utilized by corporate management to help increase the creative and productive use of ChatGPT. This scale can also help design related regulations and policies and provide a guideline for individuals for healthy and controlled use of ChatGPT. In summary, this study aims to (1) build a theoretical basis for ChatGPT and large language model studies, (2) provide practical suggestions for developers, (3) help design relevant regulations and policies, and (4) contribute to creating a healthy digital culture surrounding the use of ChatGPT by offering a novel ChatGPT literacy scale.

Digital literacy as the basis for ChatGPT literacy

To develop ChatGPT literacy, it is necessary to review the origins and concepts of digital literacy, which is closely related to AI literacy (Wang et al., 2022). Since the digital revolution, digital literacy has reshaped society and played an important role in improving competencies in various fields, such as education, banking, health, business, and legal sectors, which are closely related to quality of life (Grefen, 2021; Hunter, 2018; Munari & Susanti, 2021; Raschke, 2003; Reddy et al., 2020). Gilster (1997), in an early attempt to conceptualize digital literacy, defined the term broadly as the ability to understand and use information generated through digital sources, including AI chatbots such as ChatGPT. Later studies have extended the definition to include its

role as an essential life skill “to read, write, and otherwise deal with information using the technologies and formats” (Bawden, 2008, p. 18) and the reproduction of information and a new way of thinking that is required for the successful use of digital information (Eshet, 2012). Gilster (1997) also highlighted the importance of training dynamic thinkers with not only tangible technical skills but also intangible abilities, such as new mindsets for the digital era that are open to interactive mediums.

Such digital competencies are becoming more important because of the advent of a digital era or a new knowledge era in which digital literacy plays a key role in information acquisition, decision-making, and problem-solving in areas that are directly related to learning and quality of life (Reddy et al., 2020; Sarkar, 2012). This is particularly highlighted in the education sector, where digital literacy is considered a prerequisite for learning, because a lack of digital competencies may lead to a serious knowledge gap. This is also true outside the education sector, where digital transformation is most evident because of the need to maintain more accessible and effective learning environments. Digital literacy plays an important role in preventing the widening information gap for minorities, people with disabilities, and students from underprivileged areas (Bawden, 2008; Fu, 2013; Raschke, 2003). Moreover, digitalization blurs the boundaries between different disciplines and borders, creating “a new anthropological space” where there is an excessive amount of information from various digital sources (Lévy, 1997; Raschke, 2003). Therefore, people need to be able to find gold from gravel, detect fake information, and teach themselves through digital discernment to achieve goals in various life situations (Fu, 2013; Reddy et al., 2020).

Another reason for the increasing importance of digital literacy is the interactive nature of the information acquisition process in this era of new information. Communication and learning environments have become more interactive, transforming from a passive media environment with television and print to an active media environment with web and mobile channels, where users are expected to play active roles in the environment and consume, produce, and co-create content and meaning with others or other things, such as AI agents (Eshet, 2012; Gilster, 1997; Koc & Barut, 2016). This democratization of information and non-hierarchical, coactive network structure has given people more freedom and diverse roles and responsibilities in a digital society (Gilster, 1997; Raschke, 2003). Such expanded roles, fueled by the rapid development of digital technologies and digitalization in various fields of society, digital literacy, and digital interaction abilities to exchange and produce content and meanings, have become increasingly important (Gilster, 1997; Reddy et al., 2020; Sarkar, 2012).

Digital interactions are becoming more common and influential and have begun to resemble various real-life

social settings (Lee et al., 2023a; Namkoong et al., 2023; Park et al., 2023a, b); thus, emotional, social, and ethical factors of digital literacy have grown in importance as much as professional skills (Eshet, 2012; Huda & Hashim, 2022; Koc & Barut, 2016). Owing to the private nature of digital device environments with reduced monitoring and control, users' social and voluntary ethical behaviors and responsibilities play an important role in online settings, which are other key factors of digital literacy. In the context of ChatGPT, experts have raised various ethical considerations, such as morality concerns, information bias, misinformation issues, academic integrity concerns, and privacy concerns (Chan, 2022; Crawford et al., 2023; Wang et al., 2022).

Various taxonomies and scales for digital, AI, and new media literacy have been developed to serve the need to accurately define and measure user literacy in the digital era. Traditional studies on digital literacy have focused on the consumption of digital content; however, recently, the interactive nature of information co-creation and presumption, which involves both the production and consumption of digital content, has received attention (Gilster, 1997; Koc & Barut, 2016). For example, Koc and Barut (2016) defined digital literacy as a survival kit for the digital society; they focused on new media literacy and categorized it into four factors: functional consumption, critical consumption, functional presumption, and critical presumption. They highlighted the importance of the traditional consumption aspects of new media literacy as well as the rapidly increasing need for active production aspects of current media consumption and the technical skills required for creating and co-creating digital artifacts. Owing to the interactive nature of AI-human interaction, the importance of presumption has been further highlighted for AI literacy (Ng et al., 2021). Therefore, this study also focuses on active user engagement and the presumption of digital content in the ChatGPT context.

ChatGPT literacy scale: factor development

To extract key factors for the ChatGPT factor development, this study conducted a comprehensive literature review of digital, new media, and AI literacy. Details of each factor are presented in the following sections.

Factor 1. Technical proficiency

Traditional literature on digital literacy began with the technical proficiency of individuals in reading and writing using different technologies (Bawden, 2008). Studies on digital technical/functional skills and understanding of technology consider technical proficiency as a key factor in digital media, new media, and AI literacy (Koc & Barut, 2016; Lin

et al., 2013; Laupichler et al., 2023; Ng et al., 2021; Sarkar, 2012; Wang et al., 2022). For example, Wang et al. (2022), proposed an AI literacy scale that highlights technical skills such as “I can use AI applications or products to improve my work efficiency,” and “I can skilfully use AI applications or products to help me with my daily work” (p. 9). Koc and Barut (2016) further discussed functional consumption in the context of new media literacy and proposed the importance of the conjunction of different technologies, with scale items like “It is easy for me to make use of various media environments to reach information” (p. 842). Ng et al. (2021) also highlighted the importance of understanding the meaning of AI. In sum, common factors of the technical skills factor from the previous literature include the ability to understand and utilize the technology and other related technologies for the specific purpose of application (e.g., “I can understand how ChatGPT generates responses,” “I can train and fine-tune ChatGPT for specific purpose or application,” “I have the ability to use ChatGPT in conjunction with other tools or technologies”). Laupichler et al. (2023) highlighted the importance of the ability to understand technology in AI literacy. Based on these findings, we developed the first factor, technical proficiency with ChatGPT, and defined it as the possession of the technical skills required to understand and utilize ChatGPT.

Factor 2. Critical evaluation

Another important aspect of ChatGPT literacy is critical evaluation. Extant studies have extended the scale beyond technical skills and highlighted critical thinking skills, including the ability to evaluate and analyze the capabilities and limitations of media or technology (Bawden, 2008; Eshet, 2012; Gilster, 1997; Koc & Barut, 2016; Lin et al., 2013; Sarkar, 2012; Wang et al., 2022). For example, Wang et al. (2022, p. 9) include evaluation as one of the key factors of AI literacy (e.g., “I can evaluate the capabilities and limitations of an AI application or product after using it for a while”). Sarkar (2012) also recognized analyzing and evaluating as key techniques of information and communication technology. Koc and Barut (2016) also included critical consumption as one of the key factors of new media literacy and developed items including, “It is easy for me to make decision about the accuracy of media messages,” and “I can assess media in terms of credibility, reliability, objectivity and currency” (p. 842). Furthermore, Bawden (2008) highlighted the importance of critical thinking for digital literacy and the skills needed to judge the validity and completeness of a source to make informed decisions. Laupichler et al. (2023) also emphasized users' understanding of the possible biases created by AI systems. Based on the discussion, we defined critical thinking skills in ChatGPT literacy as the ability to

evaluate and analyze the accuracy, reliability, completeness, bias, and errors of ChatGPT's responses (e.g., "I can evaluate the accuracy of ChatGPT responses," "I can evaluate the reliability of ChatGPT's responses," and "I can recognize and explain bias in ChatGPT's responses").

Factor 3. Communication proficiency

The extant literature further highlights the importance of enhancing communication proficiency through the interactive nature of new media and the AI communication environment (Bawden, 2008; Eshet, 2012; Gilster, 1997; Koc & Barut, 2016; Ng et al., 2021). In the digital environment, it is not only important to communicate effectively with technology but also to interact and collaborate with others using the acquired information to suit a particular situation (Gilster, 1997). For example, Ng et al. (2021) include collaborative and inquiry-based learning as key factors in AI literacy. Koc and Barut (2016, p. 842) also highlighted the importance of communication and collaboration skills in the context of new media literacy (e.g., "I am able to collaborate and interact with diverse media users towards a common purpose"). Based on the discussion, we defined communication and collaboration skills as the ability to communicate effectively with ChatGPT (e.g., "I can communicate effectively with ChatGPT," "I can ask appropriate and effective questions to ChatGPT," and "I can use ChatGPT to promote collaboration or communication with others").

Factor 4. Creative application

Lin et al. (2013) categorized new media literacy into four factors (critical literacy vs. functional literacy × consuming literacy vs. presuming literacy) and defined the highest levels of literacy (critical and presuming) as the ability to create content and social impact and actively participate in the environment. The importance of discovery and creative storytelling has also been highlighted in the context of AI literacy (Ng et al., 2021). Moreover, Koc and Barut (2016, p. 842) emphasized active participation and created content production in the context of new media literacy (e.g., "I produce media content respectful to people's different ideas and private lives" and "I am good at producing opposite or alternative media contents"). Reddy et al. (2020) highlighted the importance of creative and innovative thinking skills in digital literacy. Based on this discussion, we defined creative application as the ability to generate new ideas or solutions to improve creativity and innovation skills using ChatGPT (e.g., "I can use ChatGPT to generate new ideas or solutions," and "I can improve my creativity or innovation skills using ChatGPT").

Factor 5. Ethical competence

Another important factor recognized in the extant studies is ethical competence. The importance of ethical competence is enhanced by increased active user participation and social impact (Koc & Barut, 2016; Laupichler et al., 2023; Ng et al., 2021; Wang et al., 2022). For example, Wang et al. (2022) pointed out the importance of ethical considerations in the context of AI literacy and developed the following items: "I always comply with ethical principles when using AI applications or products," and "I am never alert to privacy and information security issues when using AI applications or products" (p. 9). Koc and Barut's (2016, p. 842) new media literacy scale also includes ethical items (e.g., "It is important for me to create media contents that comply with legal and ethical rules"). Laupichler et al. (2023) also highlighted the importance of the ability to identify ethical issues and "the potential impact of AI on individuals and society" (p. 7). Based on the discussion, we defined ethical competence as the ability to identify ethical or legal issues and practice the ethical use of ChatGPT (e.g., "I can use ChatGPT ethically," and "I can identify potential ethical issues associated with the use of ChatGPT").

In summary, the literature review suggests categorizing ChatGPT literacy into five factors: (1) technical proficiency, (2) critical evaluation, (3) communication proficiency, (4) creative application, and (5) ethical competence. The following section presents the content validation process through a focus group interview with experts using the Delphi method. Furthermore, a pilot study was conducted for face validity, and a main online survey was conducted to evaluate the construct, convergent, and discriminant validity of the ChatGPT literacy scale through exploratory and confirmatory factor analyses.

Method

Item development

The development of items to measure ChatGPT literacy involved several systematic steps. First, we conducted an extensive review of the literature on digital, media, and AI literacy to gain a conceptual understanding of ChatGPT literacy. This comprehensive review helped us identify the key components and skills required for ChatGPT literacy, thus providing a solid foundation for item development. Next, we reviewed previous measures and scales related to digital and media literacy to identify potentially adaptable items. We carefully examined all relevant items and revised them to ensure that they were appropriate for measuring ChatGPT literacy in our specific context. Next, we convened a focus group comprising five experts in AI, natural language

processing, and digital and media literacy. The focus group brainstormed items representing the five factors of ChatGPT literacy: technical proficiency, critical evaluation, communication proficiency, creative application, and ethical competence. The discussions provided valuable insights into the specific skills and competencies required for ChatGPT literacy and helped ensure that the developed items were relevant, comprehensive, and aligned with the five factors of ChatGPT literacy. Based on focus group discussions and a literature review, we generated an initial pool of 34 items. We carefully worded each item to reflect the corresponding factors of ChatGPT literacy and reviewed them to ensure clarity and relevance.

Delphi validation

A Delphi survey was conducted to validate the initial 34 items designed to measure ChatGPT literacy. A panel of 10 experts, comprising professionals from diverse fields such as computer education, instructional technologies, communication research, and psychometrics, was selected for this purpose. All the experts possessed expertise in the scope and methodology of this study. They were provided with a clear definition of ChatGPT literacy factors and the item list to rate the relevance of each proposed item using a 5-point Likert scale (1 = *not at all relevant*, 5 = *very relevant*). They were also asked to identify redundant or similar concepts among the items, suggest items for deletion, and propose additional items that they felt should be included.

After collecting expert feedback and evaluation data, content validity ratios (CVR) were calculated to analyze the content validity of the 34 items. The CVR indicates the degree of agreement of the expert panel on the concepts and factors. It was computed by subtracting half of the total number of respondents ($10 / 2 = 5$) from the number of valid responses (i.e., 5 or higher on the 7-point Likert scale) and then dividing by half of the total number of respondents (Lawshe, 1975). The minimum level of agreement for CVR varied based on the number of expert panelists; as 10 panelists were included in this study, the minimum value for CVR was 0.56. Additionally, the consistency of the experts' responses to each item was evaluated using a coefficient of variation (CV). A CV of 0.5 or less (i.e., standard deviation divided by the mean) was deemed good (Okoli & Pawlowski, 2004). Therefore, we excluded six items with a CVR of 0.56 or less and a CV of 0.5 or more from the responses of the 10 expert panelists (Item 8: "I can understand and interpret natural language processing concepts and techniques related to ChatGPT;" Item 22: "I can use ChatGPT to support my decision-making process;" Item 23: "I can use ChatGPT to facilitate collaboration and communication with others;" Item 28: "I can use ChatGPT to optimize the efficiency and

productivity of my work;" Item 29: "I can use ChatGPT to achieve my learning or educational goals;" and Item 34: "I have the ability to analyze the positive or negative impact of ChatGPT on people"). As a result, the number of items was revised to 28 using the Delphi survey (see Table 1).

Pilot test

Developing a scale requires ensuring that respondents understand the items. Face validity is an approach used to evaluate whether an assessment or measure accurately measures what it is intended to measure, based on how respondents interpret it (Cohen et al., 2013). In this study, a focus group interview was conducted with 10 university students who voluntarily participated and completed the questionnaire in an empty classroom, following the survey format. All participants in this pilot test had experience using ChatGPT before participating in this test. The researchers observed the students and conducted discussions to assess their comprehension and perceptions of each item. The participants were asked to evaluate the phrasing, clarity, and logical organization of the items, which are key aspects of face validity (Anastasi & Urbina, 1997). According to the pilot test results, the unified 5-point Likert scale and items were understandable and acceptable to the participants. Analysis of their feedback revealed that all the items demonstrated good language clarity and face validity from the target audience's perspective.

Participants

This study shares the sample with a study conducted by Lee and Park (2023).

A total of 822 participants were divided into two groups: A development subsample comprising 332 participants and a cross-validation subsample comprising 490 participants recruited from an online research agency called Macromill Embrain, an online research agency in South Korea. All participants had experience using ChatGPT. Prior to participating, participants received information about the study and were asked to respond to 34 items on their ChatGPT literacy using a 5-point Likert scale (1 = *not at all*, 5 = *very much*) as part of the scale development process. Cash incentives were offered to the participants to complete the online survey. Of the total sample, 407 (49.5%) were male and 415 (50.5%) were female, with an average age of 22.7 years ($SD = 2.64$). Participants reported using ChatGPT at different frequencies: 38 (4.6%) used it as often as they could (at least once a day), 135 (16.4%) used it almost daily (5–6 times a week or more), 292 (35.5%) used it often (4–5 times a month), 216 (26.3%) used it occasionally (2–3 times a month), and 141 (17.2%) used it less than twice a month.

Table 1 Results of Delphi survey of ChatGPT literacy scale

Factor/item	Mean	SD	CVR	CV	Retain or Delete
Factor 1: Technical Proficiency (TP)					
1. I have the ability to identify and solve technical problems of ChatGPT.	6.00	0.77	0.80	0.13	Retain
2. I can train and fine-tune ChatGPT for specific purposes or applications.	5.70	1.00	0.60	0.18	Retain
3. I have the ability to use ChatGPT in conjunction with other tools or technologies.	5.80	0.75	1.00	0.13	Retain
4. I can compare and evaluate the functions of ChatGPT and other language models.	5.20	0.75	0.60	0.14	Retain
5. I can understand how ChatGPT works.	5.40	0.92	0.60	0.17	Retain
6. I can understand how ChatGPT generates responses.	5.30	0.78	0.60	0.15	Retain
7. I can adapt to technical changes in ChatGPT.	5.10	0.70	0.60	0.14	Retain
8. I can understand and interpret natural language processing concepts and techniques related to ChatGPT.	4.50	0.81	0.00	0.18	Delete
Factor 2: Critical Evaluation (CE)					
9. I can evaluate the accuracy of ChatGPT responses.	6.20	0.75	1.00	0.12	Retain
10. I can determine whether ChatGPT's response is true.	6.00	1.00	0.80	0.17	Retain
11. I can evaluate the reliability of ChatGPT's responses.	6.40	0.66	1.00	0.10	Retain
12. I can identify errors in ChatGPT's responses.	5.60	0.66	0.80	0.12	Retain
13. I can evaluate the completeness of ChatGPT's responses.	6.00	0.63	1.00	0.11	Retain
14. I can recognize and explain bias in ChatGPT's responses.	5.60	1.02	0.60	0.18	Retain
15. I can evaluate and compare the quality of different ChatGPT models or versions.	5.20	0.75	0.60	0.14	Retain
16. I can integrate ChatGPT's responses with other sources of information.	5.30	0.78	0.60	0.15	Retain
Factor 3: Communication Proficiency (CP)					
17. I can ask appropriate and effective questions to ChatGPT.	6.60	0.49	1.00	0.07	Retain
18. I can communicate effectively with ChatGPT.	6.90	0.30	1.00	0.04	Retain
19. I can use technical terms in conversations with ChatGPT.	5.20	0.75	0.60	0.14	Retain
20. I can ask accurate questions to ChatGPT using rich vocabulary.	6.30	0.46	1.00	0.07	Retain
21. I can elicit a response from ChatGPT to suit a particular situation.	6.30	0.64	1.00	0.10	Retain
22. I can use ChatGPT to support my decision-making process.	3.70	1.10	-0.20	0.30	Delete
23. I can use ChatGPT to facilitate collaboration and communication with others.	3.90	0.83	-0.40	0.21	Delete
Factor 4: Creative Application (CA)					
24. I can use ChatGPT to generate new ideas or solutions.	5.40	0.80	0.60	0.15	Retain
25. I can use ChatGPT to generate insights and trends on large datasets.	5.10	0.70	0.60	0.14	Retain
26. I can do creative writing or storytelling using ChatGPT.	5.60	0.66	0.80	0.12	Retain
27. I can improve my creativity or innovation skills using ChatGPT.	5.30	0.64	1.00	0.12	Retain
28. I can use ChatGPT to optimize the efficiency and productivity of my work.	4.70	0.64	0.20	0.14	Delete
29. I can use ChatGPT to achieve my learning or educational goals.	4.60	0.92	0.00	0.20	Delete
Factor 5: Ethical Competence (EC)					
30. I can identify potential ethical issues associated with the use of ChatGPT.	5.70	1.00	0.60	0.18	Retain
31. I can explore legal or ethical considerations related to the use of ChatGPT.	5.50	0.81	0.80	0.15	Retain
32. I can recognize potential privacy issues related to the use of ChatGPT.	5.30	0.78	0.60	0.15	Retain
33. I can use ChatGPT ethically.	6.10	0.70	1.00	0.11	Retain
34. I have the ability to analyze the positive or negative impact of ChatGPT on people.	4.90	0.70	0.40	0.14	Delete

Results

Construct validity

To evaluate the construct validity of ChatGPT literacy, we used exploratory and confirmatory factor analyses. Our study sample was divided into two groups: a development subsample comprising 332 participants and a cross-validation

subsample comprising 490 participants recruited from an online research agency. Exploratory factor analysis (EFA) was performed on the development subsample to determine the underlying factor structure of the items. Confirmatory factor analysis (CFA) was then conducted on the cross-validation subsample to validate and adjust the measurement model derived from the exploratory factor analysis. This procedure ensured that the ChatGPT literacy scale

consistently measured the same latent construct across both subsamples.

Exploratory factor analysis of ChatGPT literacy scale To assess the suitability of the data for exploratory factor analysis, we conducted a Kaiser-Meyer-Olkin (KMO) analysis and Bartlett's Test of Sphericity on the ChatGPT literacy scale. The KMO measure of sampling adequacy was 0.91, which was above the recommended threshold of 0.60. Bartlett's Test of Sphericity was also statistically significant ($\chi^2 = 3820.09$, $df = 300$, $p < .001$), indicating that the factorability and correlation matrix of the data were highly acceptable. Based on these results, we concluded that the data were appropriate for exploratory factor analysis.

Using the principal axis extraction method with varimax rotations, we conducted an exploratory factor analysis of the ChatGPT literacy scale. Initially, five factors with eigenvalues greater than 1 were identified. However, three items showed factor loadings lower than the recommended value of 0.50 (Hair et al., 2010) or similar cross-loadings with variances of less than 0.10. Therefore, we removed these items (Item 7: "I can adapt to technical changes in ChatGPT;" Item 15: "I can evaluate and compare the quality of different ChatGPT models or versions;" and Item 16: "I can integrate ChatGPT's responses with other sources of information") and conducted an exploratory factor analysis on the final five-factor model consisting of the remaining 25 items, which accounted for 61% of the total variance. After reviewing the meaning and consistency of the items according to the conceptual framework of ChatGPT literacy adopted in the previous literature review, we labeled the factors of technical proficiency (6 items), critical evaluation (6 items), communication proficiency (5 items), creative application (4 items), and ethical competence (4 items). Table 2 presents the item names, factor loadings, eigenvalues, explained variance, and Cronbach's alpha for each factor. The factor loadings ranged from 0.54 to 0.79, indicating that all items effectively measured each factor (Hair et al., 2010). All factors were internally consistent and well defined by items, as evidenced by Cronbach's alphas above the threshold of 0.70 (DeVellis, 2003) (0.85, 0.86, 0.82, 0.79, and 0.79, respectively).

Confirmatory factor analysis of ChatGPT literacy scale We conducted CFA on the ChatGPT literacy items to assess the validity of the five-factor measurement model derived from the EFA. CFA was performed using AMOS 20.0, and cross-validation subsamples were used. We used maximum likelihood estimation, which is widely used in structural equation modeling (SEM). The measurement model consisted of factors as latent constructs and items as observed variables (see Fig. 1). We evaluated the model fit using several indices,

including standardized root mean square residuals (SRMR), root mean square error of approximation (RMSEA), goodness-of-fit index (GFI), comparative fit index (CFI), and Tucker-Lewis index (TLI). We considered model fit satisfactory when GFI, CFI, and TLI values were 0.90 or higher, and SRMR and RMSEA values were 0.05 or less (Brown, 2006; Hair et al., 2010; Kline, 2005). The CFA model showed satisfactory fit indices with $\chi^2 = 1203.62$, $df = 265$, GFI = 0.90, CFI = 0.93, TLI = 0.92, SRMR = 0.05, RMSEA = 0.06. All items had statistically significant standardized parameter estimates ($p < .001$) above the recommended value of 0.50 (Hair et al., 2010). These results suggest that the ChatGPT literacy scale's measurement model, comprising five factors and 25 items, is a good fit for the observed data and has adequate construct validity.

Convergent and discriminant validity

The convergent validity of the five constructs was assessed using the composite reliability (CR) and average variance extracted (AVE) as suggested by Hair et al. (2010). Convergent validity is confirmed when CR values exceed 0.70 and AVE values surpass 0.50. Our analysis indicated that all CR values were above 0.70 and all AVE values were above or equal to 0.50, thus supporting the model's convergent validity (see Table 3).

Discriminant validity was evaluated using two distinct methods: the approach proposed by Fornell and Larcker (1981) and the heterotrait-monotrait ratio of correlations (HTMT) as suggested by Henseler et al. (2015). According to Fornell and Larcker's method, as illustrated in Table 4, discriminant validity is established if the square roots of the AVE values, shown along the diagonal axis, are greater than the corresponding inter-construct correlation coefficients. The smallest square root of AVE in our study was 0.71, which exceeded the highest correlation coefficient value of 0.58, thereby affirming the discriminant validity of our model. Additionally, the HTMT values, as presented in Table 5, were all below the threshold of 0.90. This further substantiates the discriminant validity of the measures, in line with Henseler et al.'s (2015) guidelines.

Common method bias analysis

To evaluate the presence of common method bias in our study, we employed Harman's single-factor test, executed through confirmatory factor analysis. This test operates under the premise that common method bias presents a significant concern if a singular latent factor explains over 50% of the variance in the data (Podsakoff et al., 2003). The findings from this single-factor model yielded the following: $\chi^2 = 3282.47$, $df = 275$, GFI = 0.71, CFI = 0.68, TLI = 0.65, SRMR = 0.08, RMSEA = 0.12, indicating that

Table 2 Results of exploratory factor analysis of ChatGPT literacy scale

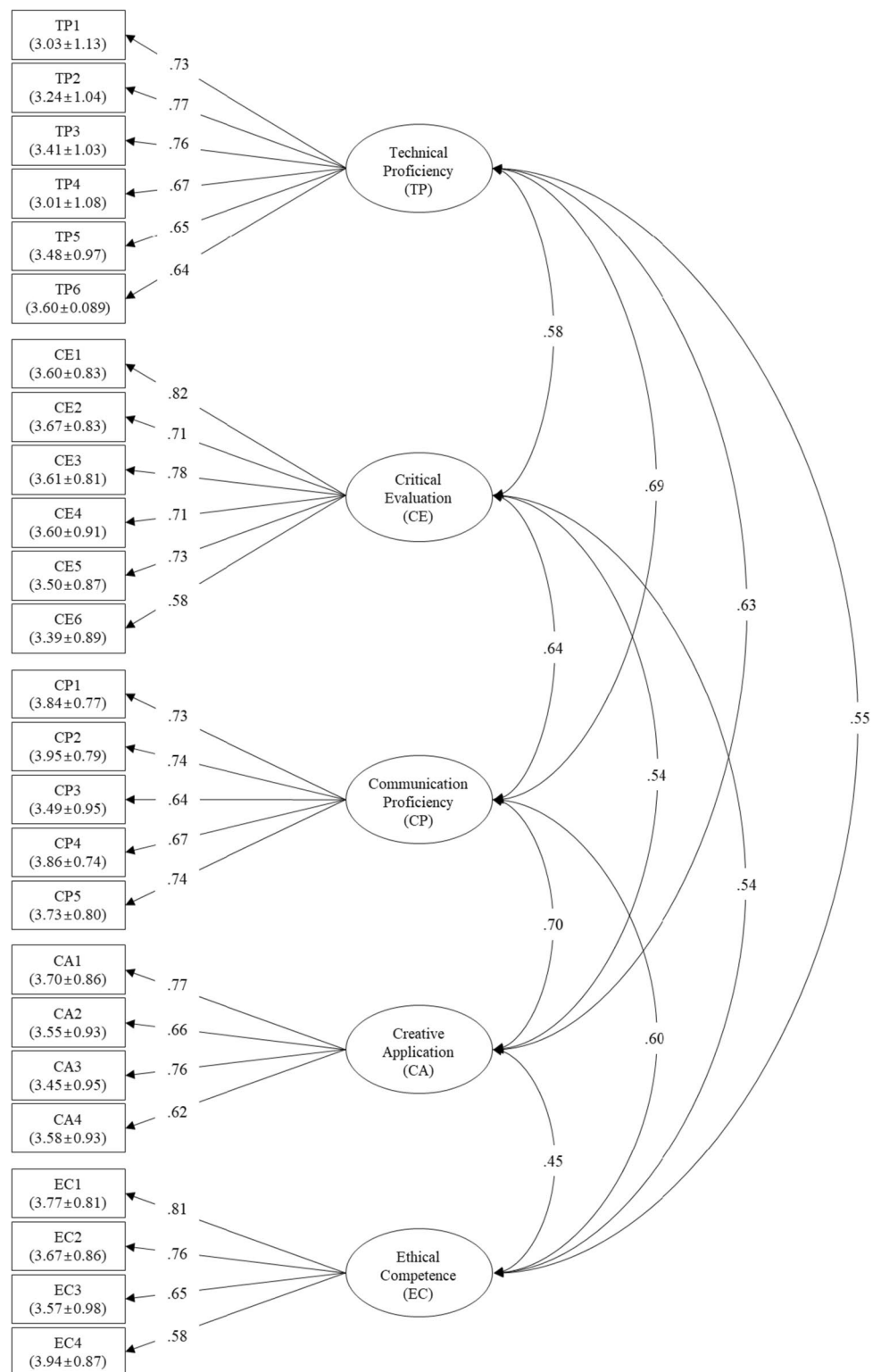
Factor/item	Factor loading				
	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5
Factor 1: Technical Proficiency (TP)					
TP1: I can train and fine-tune ChatGPT for specific purposes or applications.	0.763	0.129	0.059	0.236	0.039
TP2: I have the ability to identify and solve technical problems of ChatGPT.	0.754	0.166	0.199	0.084	0.183
TP3: I have the ability to use ChatGPT in conjunction with other tools or technologies.	0.702	0.103	0.256	0.212	0.056
TP4: I can compare and evaluate the functions of ChatGPT and other language models.	0.654	0.144	0.152	0.198	0.091
TP5: I can understand how ChatGPT generates responses.	0.614	0.237	0.175	0.044	0.258
TP6: I can understand how ChatGPT works.	0.576	0.216	0.187	0.014	0.268
Factor 2: Critical Evaluation (CE)					
CE1: I can evaluate the accuracy of ChatGPT responses.	0.125	0.786	0.177	0.151	0.148
CE2: I can determine whether ChatGPT's response is true.	0.121	0.750	0.097	0.103	0.171
CE3: I can evaluate the reliability of ChatGPT's responses.	0.143	0.742	0.261	0.066	0.127
CE4: I can identify errors in ChatGPT's responses.	0.121	0.712	0.212	0.130	0.168
CE5: I can evaluate the completeness of ChatGPT's responses.	0.254	0.679	0.154	0.182	0.058
CE6: I can recognize and explain bias in ChatGPT's responses.	0.337	0.537	-0.109	0.218	0.180
Factor 3: Communication Proficiency (CP)					
CP1: I can communicate effectively with ChatGPT.	0.208	0.150	0.754	0.128	0.156
CP2: I can ask appropriate and effective questions to ChatGPT.	0.188	0.173	0.730	0.176	0.196
CP3: I can use technical terms in conversations with ChatGPT.	0.325	0.110	0.633	0.088	0.096
CP4: I can ask accurate questions to ChatGPT using rich vocabulary.	0.062	0.263	0.582	0.286	0.211
CP5: I can elicit a response from ChatGPT to suit a particular situation.	0.271	0.237	0.541	0.362	0.169
Factor 4: Creative Application					
CA1: I can use ChatGPT to generate new ideas or solutions.	0.137	0.146	0.200	0.779	0.073
CA2: I can do creative writing or storytelling using ChatGPT.	0.172	0.123	0.106	0.729	0.071
CA3: I can use ChatGPT to generate insights and trends on large datasets.	0.238	0.115	0.106	0.705	0.313
CA4: I can improve my creativity or innovation skills using ChatGPT.	0.094	0.217	0.237	0.635	0.046
Factor 5: Ethical Competence (EC)					
EC1: I can identify potential ethical issues associated with the use of ChatGPT.	0.164	0.190	0.188	0.088	0.783
EC2: I can explore legal or ethical considerations related to the use of ChatGPT.	0.204	0.248	0.087	0.088	0.737
EC3: I can recognize potential privacy issues related to the use of ChatGPT.	0.265	0.157	0.080	0.149	0.668
EC4: I can use ChatGPT ethically.	0.019	0.079	0.333	0.136	0.654
Eigen value	1.80	8.98	1.61	1.53	1.25
Explained variance (%)	7.21	35.91	6.44	6.12	4.99
Cronbach's alpha	0.85	0.86	0.82	0.79	0.79

the model accounted for 42.4% of the total variance. Additionally, a chi-square comparison test revealed a superior fit for the five-factor model as opposed to the single-factor model ($\Delta\chi^2 = 3282.47$, $\Delta df = 10$, $p < .001$), highlighting that a single-factor model did not dominate the variance explanation. These results suggest that common method bias did not pose a significant issue in this research.

Discussion and conclusion

To meet the need for a literacy scale and facilitate the rapid uptake of ChatGPT, this study conducted an extensive literature review of digital, media, and AI literacy studies. Based on the items and factors found in the literature review, we conducted a focus group interview with experts and a Delphi

Fig. 1 Results of confirmatory factor analysis of the five-factor model of ChatGPT literacy scale. The item mean scores and standard deviations are indicated in parentheses and presented as (Mean $\pm$ SD)



study to generate an item set. Finally, we conducted a pilot study and a main online survey targeting college students to determine the factors and the final scale items through exploratory and confirmatory factor analyses, resulting in 25 final items and five factors of ChatGPT literacy: technical

proficiency, critical evaluation, communication proficiency, creative application, and ethical competence.

In this study, a novel attempt was made to develop a ChatGPT literacy scale in response to the rapid increase in ChatGPT users and ChatGPT's social impact. The scale

Table 3 Convergent validity results

ChatGPT literacy factor	Composite reliability (CR)	Average variance extracted (AVE)
Factor 1: Technical proficiency	0.86	0.50
Factor 2: Critical evaluation	0.90	0.52
Factor 3: Communication proficiency	0.88	0.50
Factor 4: Creative application	0.83	0.50
Factor 5: Ethical competence	0.84	0.50

considers both the technical and social domains of ChatGPT use, as it has created countless opportunities in various sectors that require information acquisition and creativity but also raises significant concerns about its current and latent social impact. The results of this study propose five factors of ChatGPT literacy, including technical proficiency, critical evaluation, communication proficiency, creative application, and ethical competence, and a 25-item scale to measure ChatGPT literacy, including both physical and social factors of its use. First, the technical proficiency factor was developed based on traditional scales from extant studies of digital and AI literacy (Koc & Barut, 2016; Lin et al., 2013; Laupichler et al., 2023; Ng et al., 2021; Sarkar, 2012; Wang et al., 2022). Critical evaluation is another factor commonly discussed in the context of digital literacy scales (Bawden, 2008; Eshet, 2012; Gilster, 1997; Koc & Barut, 2016; Lin et al., 2013; Sarkar, 2012; Wang et al., 2022). This study developed items by tailoring existing scales to reflect the characteristics of ChatGPT. Measuring technical proficiency and critical evaluation skills would be especially useful for educators and businesses, as there is a good chance that

these skills will become basic life and learning skills and essential communication tools in the near future.

Moreover, items in the communication proficiency and creative application factors were developed by reflecting the interactive and interconnected nature of ChatGPT to include its ability to extend human creativity and collaboration. We assume that these factors are particularly important in various collaborative situations, such as schools and companies, and help examine ChatGPT's impact on education, work, and creative activities. We believe that the ability to ask questions, have a discussion, communicate, and collaborate with AIs will become more important as non-human agents play significant parts in various fields such as prosocial behavior (Lee et al., 2023b, 2024; Park et al., 2023c) and healthcare (Park et al., 2023d). The ethical competence factor includes the ability to identify ethical issues and practice ethical presumption using ChatGPT. These measures will help solve several concerns regarding academic integrity, information biases, and misuse and abuse of large language models by helping examine the scope and extent of ethical problems and the possibility of ChatGPT extending human communication and creativity (Dwivedi et al., 2023).

Furthermore, this study is expected to become a study that establishes a basis for future studies exploring the effect of ChatGPT literacy on various outcome variables such as user satisfaction, academic achievements, creativity, and productivity. We believe this study will provide a reliable basis for studies exploring different mechanisms behind the use of ChatGPT, which is much needed amid the rapidly growing popularity of ChatGPT in almost all fields including education, healthcare, medicine, and life decisions. Moreover, this study also provides important managerial and educational implications. For example,

Table 4 Discriminant validity results using Fornell and Larcker's criterion

ChatGPT literacy factor	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5
Factor 1: Technical proficiency	0.71				
Factor 2: Critical evaluation	0.52	0.72			
Factor 3: Communication proficiency	0.58	0.54	0.71		
Factor 4: Creative application	0.51	0.47	0.56	0.71	
Factor 5: Ethical competence	0.47	0.47	0.51	0.39	0.71

The diagonal of the report shows the square roots of the Average Variance Extracted (AVE) values, while the estimates for the correlations between factors are displayed below the diagonal

Table 5 Discriminant validity results using Heterotrait-Monotrait Ratio (HTMT) correlation analysis

ChatGPT literacy factor	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5
Factor 1: Technical proficiency					
Factor 2: Critical evaluation	0.58				
Factor 3: Communication proficiency	0.68	0.63			
Factor 4: Creative application	0.61	0.54	0.69		
Factor 5: Ethical competence	0.56	0.55	0.61	0.47	

we believe the ChatGPT scale can be utilized to evaluate employees' creative and productive use of ChatGPT. Moreover, this scale can also be used to educate employees and students to improve achievements, satisfaction, productivity, and creativity. We also believe that using this scale to educate employees would not only help lower direct costs such as time and effort but also indirect costs by contributing to risk/crisis management efforts. For example, promoting the ethical use of ChatGPT can reduce the risks caused by copyright infringement and invasion of privacy.

Despite the implications of this study, some limitations must be addressed in future studies. First, the Delphi method only included Korean experts. Although South Korea is one of the leading countries in the AI transition (Song, 2022), the opinions of experts from different cultural and geographical backgrounds would improve the generalizability of the study results. Future studies should test the validity of the ChatGPT literacy scales proposed in this study in different social and cultural settings. Another limitation is the number of items included in this study. Future studies should expand the scale to include more indicators to reflect the fast-changing ChatGPT functions and applications. Moreover, we suggest that future studies utilize this scale to contribute to resolving the challenges specifically facing the education sector, such as plagiarism, copyright, and academic integrity (Cotton et al., 2023; Fu, 2013; Laupichler et al., 2023; Lin et al., 2021; Shih et al., 2021) as well as in other significantly affected areas, such as health (Biswas, 2023) and medical research (Dahmen et al., 2023). Future studies should test the validity of this scale beyond the use of ChatGPT and expand its application to other large language models.

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Data availability The data used to support the findings of this study are available from the corresponding author upon request.

Declarations

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was obtained from all individual participants included in the study.

Conflict of interest The authors declare no conflicts of interest concerning the research, authorship, or publication of this article.

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